

Fermi-Velocity-Controlled Length Scales in Solids

One Master Formula, Many Phenomena

Graduate Lecture

July 3, 2026

Outline

- 1 The Master Formula
- 2 Temperature: The Thermal Length
- 3 The Gap: BCS Coherence Length
- 4 Kondo: The Screening Cloud
- 5 Disorder: Mean Free Path and Localization
- 6 Diffusive Generalization and Thouless Energy
- 7 Summary and Limits of the Framework

Opening challenge

Energy scale		Measured length
$E_F \sim 5 \text{ eV}$	$\longrightarrow$	Ångströms
$\Delta \sim 1 \text{ meV}$	$\longrightarrow$	$\sim 200 \text{ nm}$
$k_B T \text{ at } 1 \text{ K}$	$\longrightarrow$	$\sim 1 \text{ } \mu\text{m}$
$k_B T_K \sim 0.1 \text{ meV}$	$\longrightarrow$	$\sim 1 \text{ } \mu\text{m}$

Question

What single conversion factor unifies Josephson junctions, superconductors, Kondo physics, and localization?

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$$\hbar v_F \approx 0.66 \text{ eV} \cdot \text{nm}$$

The master formula

Low-energy physics lives on the Fermi surface; linearize:

$$\varepsilon_k \simeq \hbar v_F (k - k_F) \quad \Longrightarrow \quad \text{energy window } E \leftrightarrow \text{momentum shell } \delta k = \frac{E}{\hbar v_F}$$

Fourier uncertainty: waves from a window δk stay phase-coherent over $\sim 1/\delta k$ only.

Master formula

$$L_E \sim \frac{\hbar v_F}{E}$$

Key idea

Any mechanism with energy scale E smears states within E of the Fermi surface $\Rightarrow$ imprints real space at $L_E = \hbar v_F/E$. The physics differs wildly; the length scale does not.

Three equivalent readings

- ① **Fourier / uncertainty:** shell $\delta k = E/\hbar v_F \Rightarrow$ structure of size $1/\delta k$.
(Cooper pair size, Kondo cloud)
- ② **Dephasing:** states split by E differ by $\delta k = E/\hbar v_F$; relative phase $\delta k x \sim 1$ at $x \sim \hbar v_F/E$.
(Thermal Friedel decay, SNS supercurrent)
- ③ **Dynamical:** coherence time $\tau_E = \hbar/E$; ballistic electron covers $L_E = v_F \tau_E = \hbar v_F/E$.
(Generalizes to diffusive metals: Sec. 6)

Numbers to memorize

$$\hbar v_F \approx 0.66 \text{ eV} \cdot \text{nm}, \quad k_B \times 1 \text{ K} \approx 86 \text{ } \mu\text{eV}$$

Anchor case: $E = E_F$ gives the Fermi wavelength

For a parabolic band $E_F = \frac{1}{2}\hbar v_F k_F$:

$$L_{E_F} = \frac{\hbar v_F}{E_F} = \frac{2}{k_F} \sim \lambda_F \sim \text{\AA}$$

The *shortest* length in the hierarchy — where “near the Fermi surface” breaks down.

Two roles, keep separated throughout

- k_F sets **fast oscillation periods** (π/k_F : Friedel, RKKY).
- $\hbar v_F/E$ sets **slow envelope decay lengths**.

Thermal length $L_T = \hbar v_F / 2\pi k_B T$

Set $E \sim k_B T$. More precisely: Matsubara frequencies $\omega_n = (2n + 1)\pi k_B T$; longest-lived contribution from the lowest:

$$L_T = \frac{\hbar v_F}{2\pi k_B T} \quad L_T(1 \text{ K}) \approx 1.2 \text{ } \mu\text{m}$$

Mesoscopic physics in a dilution fridge is micron-scale *for a reason*.

Two applications, which will turn out to be *the same calculation*:

- Friedel oscillations at finite T
- Supercurrent decay in a clean SNS junction

Derivation I: thermal Friedel envelope (setup)

$T = 0$: sharp Fermi edge $\Rightarrow$ endpoint-dominated integral $\Rightarrow$ power law

$$\delta n(r) \propto \frac{\cos(2k_F r + \delta_0)}{r^d}$$

$T > 0$: endpoint replaced by Fermi function. Linearize $\varepsilon = \hbar v_F q$, $q = k - k_F$:

$$\delta n(r) \propto \text{Im} \left[g(k_F) e^{2ik_F r} \underbrace{\int_{-\infty}^{\infty} dq \frac{e^{2iqr}}{e^{\beta \hbar v_F q} + 1}}_{\text{the whole game}} \right]$$

Strategy

Close the contour in the upper half q -plane and collect the poles of the Fermi factor.

Derivation I: the Matsubara pole series

Poles where $e^{\beta \hbar v_F q} = -1$:

$$q_n = \frac{i \omega_n}{\hbar v_F}, \quad \omega_n = (2n+1)\pi k_B T \quad \text{— the Matsubara frequencies!}$$

Residue = $-1/(\beta \hbar v_F)$ at each pole; geometric series resums:

$$\int_{-\infty}^{\infty} \frac{dq e^{2iqr}}{e^{\beta \hbar v_F q} + 1} = -\frac{2\pi i}{\beta \hbar v_F} \sum_{n=0}^{\infty} e^{-2\omega_n r / \hbar v_F} = -\frac{\pi i}{\beta \hbar v_F} \frac{1}{\sinh(r/L_T)}$$

Result: universal thermal envelope

$$\delta n(r; T) = \delta n(r; 0) \times \frac{x}{\sinh x}, \quad x = \frac{r}{L_T}$$

$x \ll 1$: power law survives; $x \gg 1$: $\sim 2x e^{-x}$ (exponential decay)

Asymptotics controlled by the pole nearest the real axis, $\omega_0 = \pi k_B T$.

Derivation II: clean SNS junction via Andreev phases

Supercurrent carried by Andreev pairs in the normal region of length L :

$$\text{electron: } k_e = k_F + \frac{\varepsilon}{\hbar v_F}, \quad \text{retroreflected hole: } k_h = k_F - \frac{\varepsilon}{\hbar v_F}$$

Relative phase accumulated across the junction:

$$\delta\phi(\varepsilon) = (k_e - k_h) L = \frac{2\varepsilon L}{\hbar v_F}$$

- $T = 0$: all energies coherent $\Rightarrow$ robust supercurrent.
- $T > 0$: thermal average of $e^{i\delta\phi(\varepsilon)}$ over window $\varepsilon \sim k_B T$ — **same integral** as the Friedel case, with $r \rightarrow L$. Same Matsubara poles, same $1/\sinh$:

$$I_c \propto e^{-L/L_T}, \quad L_T = \frac{\hbar v_F}{2\pi k_B T}$$

Punchline

Friedel envelope = Josephson decay: one calculation, two costumes.

Pippard's argument, then the BCS derivation

Pippard (pre-BCS): pairing involves states within Δ of the Fermi surface $\Rightarrow$ shell $\delta k = \Delta/\hbar v_F \Rightarrow$ pair size $\xi_0 \sim \hbar v_F/\Delta$.

BCS: equal-time pair amplitude

$$F(\mathbf{r}) = \sum_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad u_{\mathbf{k}} v_{\mathbf{k}} = \frac{\Delta}{2E_{\mathbf{k}}}, \quad E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}$$

Angular integral (3D) + linearization $\xi_{\mathbf{k}} = \hbar v_F(k - k_F)$:

$$F(r) \propto \frac{\sin k_F r}{r} \int_{-\infty}^{\infty} d\xi \frac{\Delta \cos(\xi r/\hbar v_F)}{\sqrt{\xi^2 + \Delta^2}} = \frac{\sin k_F r}{r} 2\Delta K_0\left(\frac{\Delta r}{\hbar v_F}\right)$$

$$\xrightarrow{r \gg \hbar v_F/\Delta} \frac{\sin k_F r}{r} e^{-r/\pi \xi_0},$$

$$\boxed{\xi_0 = \frac{\hbar v_F}{\pi \Delta}}$$

Fast k_F oscillations $\times$ slow $\hbar v_F/\Delta$ envelope: the master pattern, exactly.

Coherence length: numbers and refinements

	Δ	ξ_0
Al	0.18 meV	$\sim 1.6 \mu\text{m}$
Nb	1.5 meV	$\sim 40 \text{ nm}$

Bigger gap $\Rightarrow$ smaller pair: obvious from $L_E = \hbar v_F / E$ before any computation.

Two finite- T remarks

- 1 Near T_c : $\xi(T) \approx 0.74 \xi_0 (1 - T/T_c)^{-1/2}$ (clean GL) — $\Delta(T)$ collapses, length diverges, as the master formula demands.
- 2 Proximitized *normal* metal: decay length is L_T , not ξ_0 — no gap in N, so $k_B T$ is the only scale left.

Skill to build: which energy scale governs which region of the device?

Kondo cloud I: the exponentially small energy scale

Yosida singlet variational state; self-consistency for the binding energy:

$$1 = \frac{3J\rho}{4} \int_0^D \frac{d\epsilon}{\epsilon + \Delta_b} \quad \Rightarrow \quad \Delta_b \approx D e^{-4/(3J\rho)} \sim k_B T_K$$

- Infrared logarithm from integrating $1/\epsilon$ down to a *sharp Fermi edge* — same mechanism as the Cooper log.
- Bound orbital built from amplitudes $\alpha_k \propto (\epsilon_k + \Delta_b)^{-1}$: a shell of width $k_B T_K$ above E_F .

Kondo cloud II: real-space profile

Fourier transform with linearized dispersion (s-wave):

$$\phi(r) \propto \frac{1}{r} \operatorname{Im} \int_0^D d\epsilon \frac{e^{i(k_F + \epsilon/\hbar v_F)r}}{\epsilon + k_B T_K}$$

Pole at distance $k_B T_K$ from the edge $\Rightarrow$ envelope scale

$$\boxed{\xi_K = \frac{\hbar v_F}{k_B T_K}} \quad |\phi(r)|^2 \sim \begin{cases} \frac{\sin^2 k_F r}{r^2} \ln^2 \frac{\xi_K}{r}, & r \ll \xi_K \\ \frac{\cos^2 k_F r}{r^2} \frac{\xi_K^2}{r^2}, & r \gg \xi_K \end{cases}$$

The shock

T_K exponentially small $\Rightarrow \xi_K$ exponentially large: $T_K \sim 1$ K gives $\xi_K \sim$ microns — bigger than the device! Eluded detection for 50 years; observed in a quantum-dot interferometer (Borzenets *et al.*, Nature 2020), confirming the $\hbar v_F/k_B T_K$ scaling.

Disorder: $E = \hbar/\tau$ gives the mean free path

Elastic scattering time $\tau \Rightarrow$ level broadening $\hbar/\tau$:

$$L_{\hbar/\tau} = \frac{\hbar v_F}{\hbar/\tau} = v_F \tau = \ell$$

Strict 1D derivation sketch. Transfer matrices ($\det T_n = 1$) + Furstenberg $\Rightarrow$ positive Lyapunov exponent $\gamma = 1/\xi_{\text{loc}}$; weak disorder:

$$\xi_{\text{loc}} = \frac{8t^2 \sin^2(ka)}{\langle \epsilon_n^2 \rangle}$$

Restore $\hbar v(E) = 2ta \sin(ka)$ and use the 1D Born rate $\hbar/\tau = 2\pi \langle \epsilon^2 \rangle \rho_{1D} a$, $\rho_{1D} = 1/\pi \hbar v$:

$$\xi_{\text{loc}} \simeq 4\ell = 4v_F \tau$$

Same master formula, with the disorder-induced energy scale.

Localization: dimensionality dressing (and honesty)

$$\xi_{\text{loc}} \sim \ell \text{ (1D)}, \quad \xi_{\text{loc}} \sim N_{\text{ch}} \ell \text{ (quasi-1D)}, \quad \xi_{\text{loc}} \sim \ell e^{\pi k_{\text{F}} \ell / 2} \text{ (2D)}$$

Honesty clause

The master formula supplies the *brick* ℓ . The 2D exponential and the channel-number factor come from interference physics (weak localization, scaling theory) — genuinely beyond dimensional analysis.

The framework tells you the bricks, not always the architecture.

Dirty metals: $L_E \rightarrow \sqrt{\hbar D/E}$

Dynamical reading (R3): distance traveled during $\tau_E = \hbar/E$. When $L_E \gg \ell$, motion is diffusive, $D = v_F \ell/d$:

$$L = v_F \tau_E \longrightarrow L = \sqrt{D \tau_E} \quad \Longrightarrow \quad \boxed{\frac{\hbar v_F}{E} \longrightarrow \sqrt{\frac{\hbar D}{E}}}$$

Every ballistic length has a dirty partner:

$$\xi_0 = \frac{\hbar v_F}{\pi \Delta} \rightarrow \sqrt{\frac{\hbar D}{\Delta}} \sim \sqrt{\xi_0 \ell}, \quad L_T \rightarrow \sqrt{\frac{\hbar D}{2\pi k_B T}}$$

v_F hasn't disappeared — it hides inside D .

Unified statement (both regimes)

The length associated with E is the distance a Fermi-surface electron propagates during the coherence time $\hbar/E$.

The Thouless energy: the inverse dictionary

Given a sample of size L , the associated energy:

$$\bar{\vartheta} = \frac{\hbar v_F}{L} \text{ (ballistic),} \quad \bar{\vartheta} = \frac{\hbar D}{L^2} \text{ (diffusive)}$$

Application: short vs. long Josephson junction

$$\Delta \lesssim \bar{\vartheta} \iff \xi \gtrsim L$$

Classifying the device = comparing two entries of the dictionary.

Goal for students: translate *freely in both directions* between energies and lengths.

Summary table: the hierarchy

Energy E	Length $\hbar v_F/E$	Physics	Size
E_F	$\lambda_F \sim 1/k_F$	Friedel/RKKY period	$\text{\AA}$
$\hbar/\tau$	$\ell = v_F \tau$	mean free path	nm- μm
Δ	$\xi_0 = \hbar v_F/\pi \Delta$	Cooper pair size	10 nm- μm
$2\pi k_B T$	L_T	Friedel/SNS decay	μm @ 1 K
$k_B T_K$	ξ_K	Kondo cloud	μm
$\bar{\delta}$	L (by def.)	size $\rightarrow$ energy	—

Top to bottom: energies large $\rightarrow$ small, lengths short $\rightarrow$ long.

Why quasiparticles work at all

The nesting $\lambda_F \ll \ell \ll \xi_0, L_T, \xi_K$ is what makes the quasiparticle description self-consistent.

Where the framework fails

The construction needs a sharp Fermi surface and finite v_F . Three frontiers break it:

- **Flat bands** (magic-angle TBG): $v_F \rightarrow 0$ by design. All lengths collapse to the lattice scale; interactions win; correlated insulators and superconductivity emerge. Flat-band engineering = deliberate demolition of this lecture.
- **Non-Fermi liquids**: no quasiparticles; ω/T scaling. The Planckian *time* $\hbar/k_B T$ survives, but there is no clean v_F to convert it into a length.
- **Dirac materials**: no Fermi surface at neutrality, yet the velocity survives as an emergent “speed of light.” Master formula still works with E measured from the Dirac point.

Problems

- 1 Derive the $x/\sinh x$ thermal envelope of 1D Friedel oscillations from the finite- T Lindhard function; identify the controlling Matsubara pole.
- 2 Clean SNS junction: compute $\delta\phi(\varepsilon)$ and show $I_c \propto e^{-L/L_T}$. Redo for a diffusive junction using $\tilde{D} = \hbar D/L^2$.
- 3 Estimate ξ_K for $T_K = 0.5$ K, $v_F = 1.5 \times 10^6$ m/s; compare with a quantum-dot device. What does “cloud bigger than the device” mean physically?
- 4 (Open-ended) Quadratic dispersion near E_F : how does the energy-length dictionary change? Which table entries are modified?

One number,

$$\hbar v_F \approx 0.66 \text{ eV} \cdot \text{nm},$$

converts every energy scale in a metal into a length:

$$L_E \sim \frac{\hbar v_F}{E} \quad (\text{ballistic}), \quad L_E \sim \sqrt{\frac{\hbar D}{E}} \quad (\text{diffusive}).$$

Josephson junctions, superconductors, Kondo screening, and localization: one subject, not a list of phenomena.